

① Design & configuration of star, bus, ring topologies.

② Data link layer traffic simulation using P.T Analysis of APR

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Model Practicals

Sub: CSA0711 Computer Networks

① Aim:

To design and configure star, bus, and ring topology

Materials required:

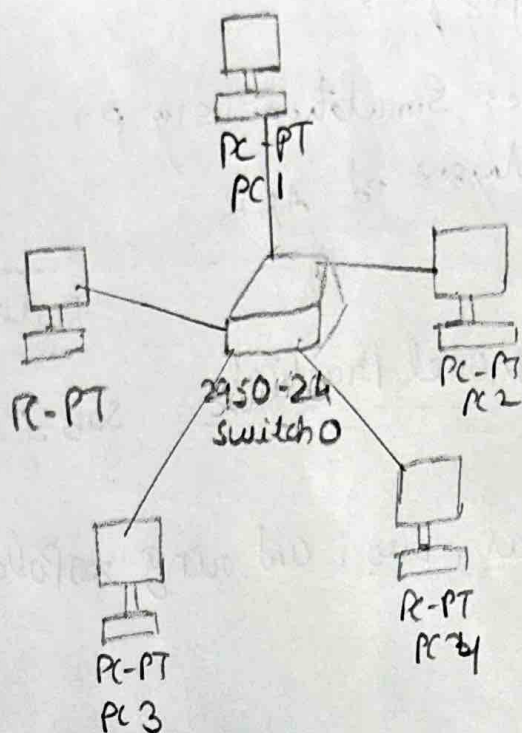
- i) Switch 2950-24
- ii) PC
- iii) wire/cable

o/p
verified

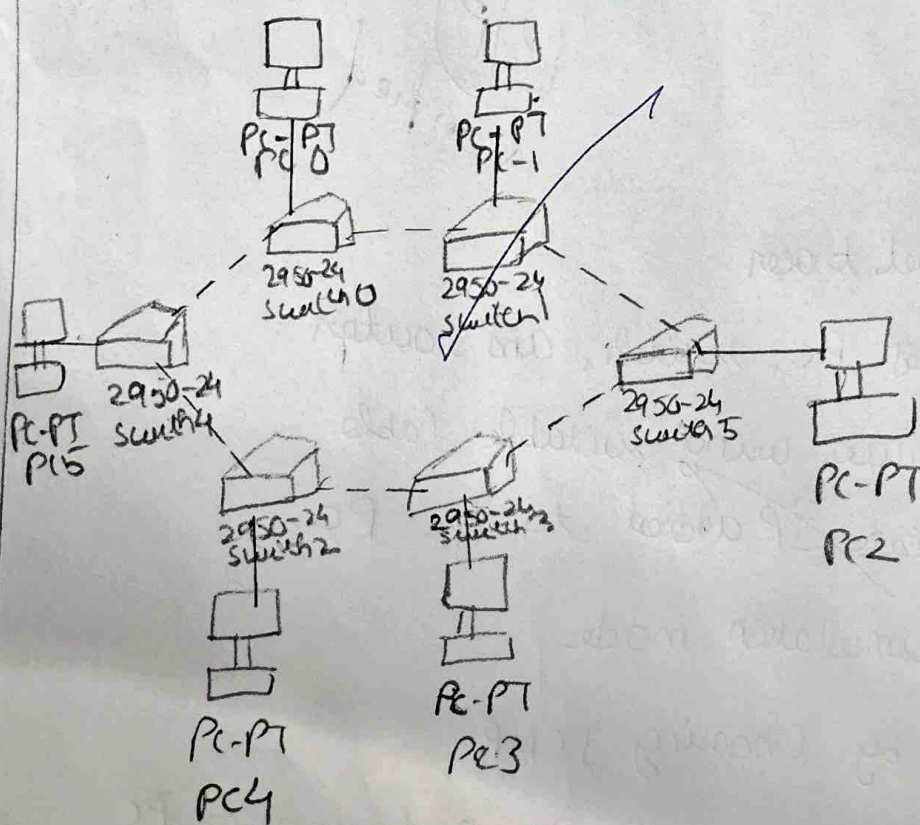
Procedure:

- 1) Open Cisco Packet Tracer
- 2) Add the required PC, switch, and router
- 3) Connect the devices using suitable cable
- 4) Assign appropriate IP address to the PC
- 5) Then go to simulation mode
- 6) Edit filters by choosing ICMP
- 7) Add simple PDU on one PC and another PC
- 8) Verify the successful communication between them

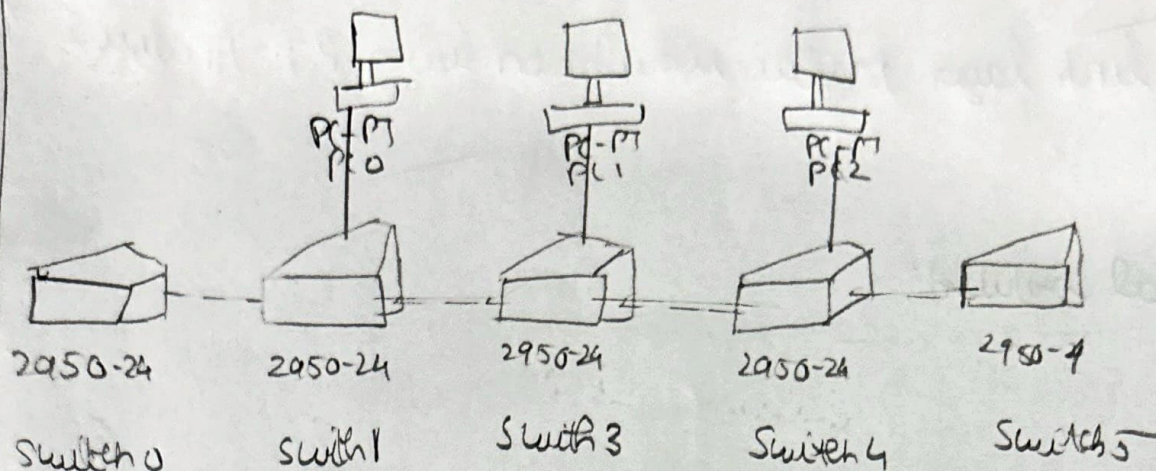
Star Topology



Ring topology:



Star topology:



Output:

Star topology successfully verified

Bus topology successfully verified

Ring topology successfully verified

Result:

Thus star, bus, ring topology are successfully designed and configured.

② Aim:

Data link layer traffic simulation using PT Analysis

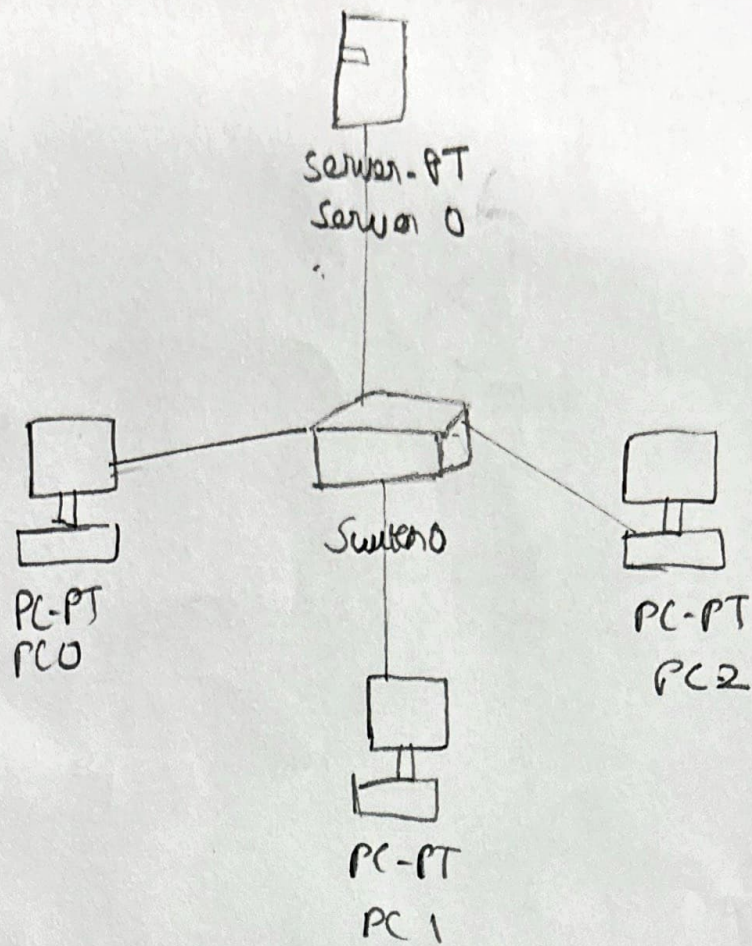
ARR

Material required:

- i) PC
- ii) server
- iii) cisco Packet Tracer
- iv) Cables

Procedure:

- i) Create the topology with PC0, PC1, server1, and switch0
- ii) Assign IP addresses in the same network:
- iii) Switch to simulation mode and select ARP and ICMP
- iv) In PC0, open command prompt and type ping 192.168.10.4
- v) Click Capture / Forward to observe the ARP request and ARP reply
- vi) Observe the subsequent ICMP data packets between PC0 and server1
- vii) Verifies the ARP table using arp -a



Result:

Thus data traffic was successfully simulated using ARP

[Signature]